FAD

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Identity Sheet

Property	Value
Name	Flavin Adenine Dinucleotide
Other names	FAD, flavin coenzyme
Chemical formula	$C_{27}H_{33}N_9O_{15}P_2$
Molecular weight	785.55 g/mol
CAS number	146-14-5
PubChem CID	643975
Appearance	Yellow-orange crystalline; intensely fluorescent (oxidised form)
Solubility	Freely soluble in water

Structural formula:

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Isoalloxazine ring–Ribitol–PO4–PO4–Ribose–Adenine
(tricyclic flavin chromophore is the fluorescent moiety)
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FAD is a redox-active coenzyme derived from riboflavin (vitamin B2). The **oxidised form (FAD)** is fluorescent (Ex 450 nm / Em 525 nm), while the reduced form ($FADH_2$) is not. FAD is a prosthetic group for flavoproteins involved in the electron transport chain (Complex II), fatty acid oxidation, and amino acid metabolism. In urine, FAD and its precursor riboflavin contribute to the characteristic yellow-green fluorescence.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Riboflavin (vitamin B2)	$C_{17}H_{20}N_4O_6$	376.36	Precursor of FAD/FMN; similar fluorescence; the dominant urinary fluorophore
FMN (flavin mononucleotide)	$C_{17}H_{21}N_4O_9P$	456.34	Intermediate between riboflavin and FAD; also fluorescent
NADH	$C_{21}H_{29}N_7O_{14}P_2$	665.44	Different coenzyme; fluorescence at 340/460 nm (not 450/525 nm)
$FADH_2$ (reduced FAD)	$C_{27}H_{35}N_9O_{15}P_2$	787.56	Reduced form; NOT fluorescent

Medical Information

Origin

Endogenous FAD is synthesised from riboflavin (vitamin B2) in two steps: riboflavin -> FMN (riboflavin kinase) -> FAD (FAD synthetase/FMN adenylyltransferase). Most FAD is tightly bound to flavoproteins (~75 flavoproteins in humans). Free FAD concentration in cells is very low.

Exogenous FAD is not obtained directly from diet. Its precursor, **riboflavin (B2)**, is the dietary source: dairy, eggs, lean meats, green vegetables, fortified cereals. RDA: 1.1-1.3 mg/day.

Primary & Secondary Biological Roles

Primary role: - **Electron carrier:** FAD/$FADH_2$ is a prosthetic group for succinate dehydrogenase (Complex II), acyl-CoA dehydrogenases, xanthine oxidase, monoamine oxidase, D-amino acid oxidase, and ~75 other flavoproteins.

Secondary roles: - **Antioxidant systems:** Glutathione reductase requires FAD. - **Metabolic imaging marker:** FAD/NADH fluorescence ratio ("optical redox ratio") used in research for tissue metabolic state assessment.

Catabolism and Elimination Pathway

- FAD-bound flavoproteins are degraded by lysosomal proteases; FAD is hydrolysed to FMN and then riboflavin.
- Riboflavin is the primary urinary excretion product (freely filtered, partially reabsorbed via tubular secretion).
- Urinary riboflavin (not FAD per se) is the dominant yellow-green fluorescent species.
- Excess riboflavin from diet/supplements is rapidly excreted, producing bright yellow urine.

Expression in Humans

Normal Levels

Compartment	Reference Range
Plasma riboflavin	5-50 nmol/L
Erythrocyte FAD	300-600 nmol/L (functional status marker)
Urinary riboflavin	120-400 ug/day (diet-dependent)
Urinary FAD	Trace (most excreted as riboflavin, not intact FAD)

Factors Influencing Levels **Increased urinary riboflavin/FAD fluorescence:** - Riboflavin supplements (bright yellow urine) - High B2 dietary intake - B-complex vitamins - Increased protein catabolism

Decreased: - Riboflavin deficiency (ariboflavinosis) - Malabsorption - Alcoholism

Associated Pathologies

Condition	FAD/Riboflavin Pattern	Key Symptoms
Ariboflavinosis (B2 deficiency)	Low urinary riboflavin, low erythrocyte FAD	Angular cheilitis, glossitis, anaemia
Multiple acyl-CoA dehydrogenase deficiency (MADD)	FAD-responsive variant	Metabolic acidosis, hypoglycaemia, myopathy

Presence in Urine

Should it be normally present? Yes – riboflavin (B2) is normally excreted in urine. Its fluorescence (and that of minor FMN/FAD) gives urine its yellow-green fluorescence under UV light.

Normal urinary levels: Riboflavin: 120-400 ug/day. Intact FAD: trace amounts.

Form in urine: Primarily as free **riboflavin**; minor amounts of FMN and FAD.

Solubility: Freely soluble.

Detection in Urine

Available Clinical Assays

1. **HPLC with fluorescence detection:**
   - **Principle:** Reversed-phase HPLC separates riboflavin, FMN, and FAD.
   - **Detection:** Fluorescence at **Ex 450 nm / Em 520-525 nm**.
   - **LOD:** ~1-5 nmol/L.
   - **Advantages:** Specific, separates FAD from riboflavin and FMN.
2. **Erythrocyte glutathione reductase activity coefficient (EGRac):**
   - **Principle:** Functional test for riboflavin/FAD status; ratio of enzyme activity with/without added FAD.
   - **Detection:** NADPH consumption at 340 nm.
   - **EGRac >1.4** indicates deficiency.
   - **Not a direct urine assay** but the clinical gold standard for B2 status.

Optimal Urine Type for Measurement

- **24-hour urine** for total riboflavin excretion.
- Protect from light (riboflavin is photolabile).
- Store at 4 C or freeze.

Actual Gold Standard

HPLC-fluorescence (Ex 450 / Em 520 nm) for urinary riboflavin/FAD. EGRac for functional status.

Interferences in Measurement

Interference	Effect	Mechanism
Light exposure	False decrease	Photodecomposition of flavins
Tetracyclines	Fluorescence interference	Similar emission wavelength
Other urinary fluorophores	Spectral overlap	Porphyrins, 4-pyridoxic acid

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis:** FAD and riboflavin absorb at **267 nm** (epsilon ~32,000) and **373 nm** (epsilon ~9,300) and **450 nm** (epsilon ~11,300 M-1cm-1). The 450 nm band gives the yellow colour.
  - LOD: ~0.1 uM by direct UV-Vis.
  - Sample prep: None.

Fluorescence Detection This is the definitive detection modality for FAD/riboflavin:

- **Direct fluorescence: Ex 450 nm / Em 520-530 nm** (green fluorescence). Quantum yield ~25% (riboflavin), ~3% (FAD, quenched by adenine).
  - LOD: ~1 nM (riboflavin), ~10 nM (FAD).
  - Urine can be measured directly with appropriate dilution.

- **Excitation-emission matrix (EEM) fluorescence:** Simultaneous multi-excitation scanning to resolve FAD, riboflavin, FMN, and other fluorophores.
  - LOD: ~5 nM with chemometric deconvolution.
  - Sample prep: None (direct) or dilution.
- **Time-resolved fluorescence:** FAD lifetime (~2.3 ns free, ~0.3 ns quenched in proteins) distinguishes it from riboflavin (~4.7 ns).

Raman Detection

Peak (cm-1)	Assignment
~1350	Isoalloxazine ring (CN stretch)
~1500	Isoalloxazine C=N stretch
~1580	Isoalloxazine C=C stretch
~1630	C=O stretch

- Conventional Raman: LOD ~100 uM.
- **Resonance Raman (Ex 458/514 nm):** Enhanced by factor of ~10^3 due to isoalloxazine absorption. LOD: ~0.1-1 uM.
- SERS: LOD ~1-10 nM with Ag substrates.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1650	C=O stretch	Isoalloxazine
~1580	C=C/C=N stretch	Ring
~1540	NH bending	Amide
~1080	PO4 stretch	Phosphodiester

- Not practical for urinary FAD detection (trace levels).

Voltammetry Detection

- FAD undergoes two-electron two-proton reduction: FAD + 2H+ + 2e- -> FADH2.
- **Reduction potential:** ~-0.22 V vs SHE at pH 7 (~-0.46 V vs Ag/AgCl).
- LOD: ~0.1 uM on carbon electrodes.
- Used in biosensor design (FAD-mediated glucose oxidase sensors).

Other Detection Technologies

1. **LC-MS/MS:** Definitive identification. LOD: ~0.1 nM. Sample prep: protein precipitation.
2. **Microbiological assay (Lactobacillus rhamnosus growth):** Traditional B2 assay. LOD: ~1 nM. Time: 16-24 h.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
HPLC-fluorescence	~1-5 nM	Ex 450/Em 520 nm	None/dilute	Specific fractionation
Direct fluorescence	~1-10 nM	Ex 450/Em 525 nm	None	Simple, fast
UV-Vis	~0.1 uM	450 nm	None	Direct
Resonance Raman	~0.1 uM	Ex 458 nm	None	Specific
SERS	~1-10 nM	1350 cm-1	Ag nanoparticles	Ultra-sensitive
Voltammetry	~0.1 uM	-0.46 V	Buffer	Real-time
LC-MS/MS	~0.1 nM	m/z 786->348	Extraction	Definitive

References

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